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75

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TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 3

CLOSED BOOK, TIME: 100 MINUTES

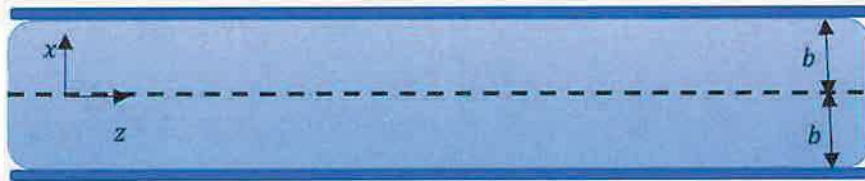
- 40/40
1. Two solid spheres of identical material but different diameters R_1 and R_2 are allowed to sink freely in the same liquid. Calculate the ratio of their terminal velocities $\frac{v_2}{v_1}$, assuming $\frac{R_2}{R_1} = 4$ and that the friction factor for both is constant $f = 0.44$.

Hint: Drag force of the liquid of density ρ on the sphere of radius R and velocity v is $AfK = (\pi R^2) f \left(\frac{1}{2} \rho v^2 \right)$.

The buoyant force of the liquid on the sphere is $\frac{4}{3} \pi R^3 \rho g$.

$$F_K = \left(\frac{4}{3} \pi R^3 \rho g - \frac{4}{3} \pi R^3 \rho g \right) = \pi R^2 \left(\frac{1}{2} \rho v^2 \right)$$

2. Consider a Newtonian fluid with constant viscosity μ and constant density ρ flowing between two large horizontal parallel plates, a distance $2b$ apart. For $t \leq 0$, flow is driven by a pressure gradient in z -direction according to the velocity profile: $v_z(x, t \leq 0) = v_{z,max} \left[1 - \left(\frac{x}{b} \right)^2 \right]$. At $t = 0$ the pressure gradient is turned off, after which the flow evolves in time towards stagnation.



- 10/5
- 2.1. Simplify the equation of motion to derive a partial differential equation (PDE) for the velocity profile for $t > 0$.

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

- 10/5
- 2.2. Rewrite the PDE in terms of dimensionless variables $\phi = \frac{v_z}{v_{z,max}}$, $\xi = \frac{x}{b}$, and $\tau = \frac{\mu}{\rho b^2} t$.

- 5/5
- 2.3. Write down one initial condition and two boundary conditions for the dimensionless PDE.

- 2.4. Solve the dimensionless PDE, knowing the solution may be written as an infinite sum of cosine functions $\phi(\xi, \tau) = \sum_{n=1}^{\infty} f_n(\tau) \cos\left(\frac{2n-1}{2} \pi \xi\right)$. Evaluate the integration constants using the orthogonality equation:

$$\int_{-1}^1 d\xi \cos(n\pi\xi) \cos(m\pi\xi) = \delta_{nm}$$

Hint: $\int_{-1}^1 \left[(1 - \xi^2) \cos\left(\frac{2n-1}{2} \pi \xi\right) \right] d\xi = -(-1)^n \frac{32}{[\pi(2n-1)]^3}$

$$\phi(\xi, \tau=0) = 0 = f_n(\tau=0) \geq 0$$

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$$\begin{cases} \frac{1}{2\mu} \left(\frac{\partial P}{\partial z} \right) b^2 + c_1 b + c_2 = 0 \\ + \frac{1}{2\mu} \left(\frac{\partial P}{\partial z} \right) b^2 - c_1 b + c_2 = 0 \end{cases}$$

$$\frac{1}{\mu} \left(\frac{\partial P}{\partial z} \right) b^2 + 2c_2 = 0 \quad c_2 = -\frac{1}{2\mu} \left(\frac{\partial P}{\partial z} \right) b^2$$

$$\frac{1}{2\mu} \left(\frac{\partial P}{\partial z} \right) b^2 + c_1 b - \frac{1}{2\mu} \left(\frac{\partial P}{\partial z} \right) b^2 = 0 \Rightarrow c_1 = 0$$

$$v_z = -\frac{1}{2\mu} \left(\frac{\partial P}{\partial z} \right) b^2 \left(1 - \left(\frac{r}{b} \right)^2 \right) \quad v_{z, \max} = -\frac{1}{2\mu} \frac{\partial P}{\partial z} b^2$$

$$\mu \left(-\frac{1}{2\mu} \frac{\partial P}{\partial z} b^2 \right) \left(\frac{r}{b^2} \right) \frac{\partial \phi}{\partial z} = \mu \left(-\frac{1}{2\mu} \frac{\partial P}{\partial z} b^2 \right) \left(\frac{1}{b^2} \right) \frac{\partial^2 \phi}{\partial z^2} - \frac{\partial P}{\partial z}$$

$$-\frac{1}{2} \frac{\partial P}{\partial z} \frac{\partial \phi}{\partial z} = -\frac{1}{2} \frac{\partial P}{\partial z} \frac{\partial^2 \phi}{\partial z^2} - \frac{\partial P}{\partial z} \quad || \cdot -2 \cdot \frac{\partial z}{\partial P}$$

$$\boxed{\frac{\partial \phi}{\partial z} = \frac{\partial^2 \phi}{\partial z^2} + 2}$$

$$\begin{aligned} 2.3 \quad v_z(r=b, t) &= 0 \\ v_z(r=-b, t) &= 0 \\ v_z(r, t=0) &= 0 \end{aligned}$$

$$\rightarrow \begin{cases} v_z(\xi=1, \tau) = 0 \\ v_z(\xi=-1, \tau) = 0 \\ v_z(\xi, \tau=0) = 0 \end{cases} \begin{matrix} \text{boundary condition} \\ \text{initial condition} \end{matrix}$$

$$2.4 \quad \phi(\xi, \tau) = \sum_{n=1}^{\infty} f_n(\tau) \cos\left(\frac{2n-1}{2} \pi \xi\right)$$

$$\frac{\partial \phi}{\partial \tau} = \sum_{n=1}^{\infty} \frac{\partial f_n}{\partial \tau} \cos\left(\frac{2n-1}{2} \pi \xi\right)$$

$$\frac{\partial \phi}{\partial \xi} = \sum_{n=1}^{\infty} f_n(\tau) \left(-\sin\left(\frac{2n-1}{2} \pi \xi\right) \right) \left(\frac{2n-1}{2} \pi \right)$$

$$\frac{\partial^2 \phi}{\partial \xi^2} = \sum_{n=1}^{\infty} f_n(\tau) \left(\frac{2n-1}{2} \pi \right)^2 \left(-\cos\left(\frac{2n-1}{2} \pi \xi\right) \right)$$